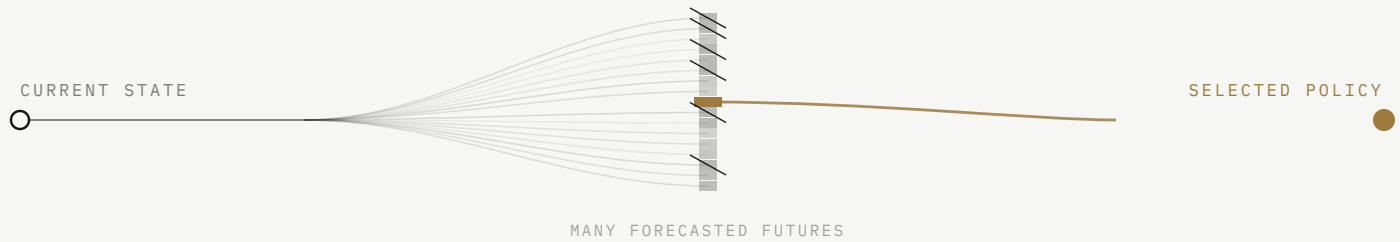


— THE UNIT OF OPTIMIZATION

Everyone optimizes the next decision. Aurelius optimizes the fleet's trajectory.

FORK • SIMULATE UNDER CONSTRAINTS • SELECT ONE

FIG. ONE PLANNING CYCLE



WHAT IT IS

It is not a forecaster and not a scheduler replacement. It is a **supervisory layer** that sits above Kubernetes, Slurm, or a serving router and never touches payloads. Each control period it forks the current fleet state into many forecasted futures, applies coupled policies across the controls the stack already exposes, advances each future under SLA, capacity, and power constraints, kills the ones that violate, and returns one policy for the existing stack to execute.

WHAT WE FOUND

The finding is narrow, and that is why it holds. We held the forecasts, the objective, the simulator, and the constraints constant, and changed only the search. A single-lever search underperformed and broke SLA. A **coupled search over forecasted futures** turned it strongly positive and, where exhaustive enumeration was tractable, matched it exactly, at zero measured regret. Under uncapped production load the same architecture separates from a fixed policy by several fold.

— If that generalizes, it is a different design axis, not a tuning win.

The full technical report carries the numbers, the production-class baseline, and the method. This page is only the idea.